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Toward Automated Data Extraction: A Cross-Sectional Pilot Study of the Structure of Tabular Data in Clinical Comparative Literature

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Abstract

Background: Systematic reviews depend on time-consuming extraction of data from PDFs of underlying studies. To date, automation efforts have focused on extracting from the text, and no approach has yet succeeded in fully automating ingestion of quantitative evidence. However, the majority of relevant data is generally presented in tables, and tabular structure is more amenable to automated extraction than free-text.

Objective: The purpose of this survey is to classify the structure and format of descriptive statistics reported in tables in the comparative medical literature.

Methods: We sampled 100 published randomized controlled trials (RCTs) from the year 2019 from PubMed; these results were imported to the AutoLit platform. Studies were excluded if they were non-clinical, non-comparative, not in English, protocol-only, or not available in full text. In AutoLit, tables reporting baseline or outcome data in all studies were characterized based on reporting practices. Measurement context, meaning the structure in which the interventions of interest, patient arm breakdown, measurement timepoints, and data element descriptions were presented, was classified based on the number of contextual pieces and metadata reported. Then, the statistic formats for reported metrics (specific instances of reporting of data elements) were classified by location and broken down into reporting strategies for continuous, dichotomous, and categorical metrics.

Results: We included 78 of 100 studies, one of which (1.3%) did not report data elements in tables. The remaining 77 studies reported baseline and outcome data in 174 tables, and 97% of these tables broke down reporting by patient arms. Fifteen structures were found for the reporting of measurement context, which were broadly grouped into: 1x1 Contexts, where two pieces of context are reported total (e.g. “arms in columns, data elements in rows); 2x1 Contexts, where two pieces of context are given on row headers (e.g. timepoints in columns, arms nested in data elements on rows); 1x2 Contexts, where two pieces of context are given on column headers. 1x1 contexts were present in 57% of tables, compared to 20% for 2x1 and 15% for 1x2 (8% used unique/other stratification). Statistic formats were reported in the headers or descriptions of 84% of studies.

Conclusions: In this pilot survey, we found a high density of information in tables, but with major heterogeneity in presentation of measurement context. The highest-density studies reported both baseline and outcome measures in tables, with arm-level breakout, intervention labels and arm sizes present, and reported both the statistic formats and units. The measurement context formats presented here, broadly classified into three classes that cover 92% of studies, form a basis for understanding the frequency of different reporting styles, supporting automated detection of data format for extraction of metrics.

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Abstract

Text 5
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Conclusions: In this cross-sectional pilot study, we found a high density of information in tables, but with major heterogeneity in presentation of measurement context. The highest-density studies reported both baseline and outcome measures in tables, with arm-level breakout, intervention labels and arm sizes present, and reported both the statistic formats and units. The measurement context formats presented here, broadly classified into three classes that cover 92% of studies, form a basis for understanding the frequency of different reporting styles, supporting automated detection of data format for extraction of metrics.

Text 6
Keywords: Table structure, Systematic review, Automated data extraction, Data reporting conventions, Clinical comparative data, Data elements, Statistic formats.

Introduction

Extracting data for systematic review

Systematic review and meta-analysis (SR/MA) of high-quality studies are essential to clinical decision-making [1], guidelines [2], and evidence-based adoption and approval of therapies [3]. Quantitative data extraction is an essential task in the SR/MA process, during which researchers gather patient characteristics, interventions, and outcomes of interest in a common format to support summarization and statistical analysis. Current practice for data extraction is manual review of published manuscripts of studies, with hand-entry of data into a spreadsheet or review software [4]. The manual, work-intensive nature of this task contributes to the high cost in time and money of reviewing the clinical literature.

The time investment and costs of SR/MA—which can reach 16 months and $141,000 [5] in labor to complete a single review—are a major limiting factor in the synthesis of scientific evidence. The task of data extraction from published comparative studies typically demands 20% of the total SR/MA time and is subject to high accuracy standards [6, 7]. This has led to calls for both improved software systems for SR/MA and the automation of the data extraction process, but according to a systematic review of SR/MA extraction automation projects, "no unified information extraction framework [has been] tailored to the systematic review process...[automation] techniques have not been fully utilized to fully or even partially automate the data extraction step of systematic review" [8].

Systematic review workflow software platforms

Despite the fact that automated data extraction for SR/MA has yet to be achieved, several web-based software options currently support part or all of the workflow of a review [9], establishing a systematic approach on which automated data extraction can be modelled. The authors have built an SR/MA workflow software platform (AutoLit, Nested Knowledge, MN) [10] for performing and presenting systematic reviews and meta-analyses. Its data extraction functions are user-driven and focused on extracting descriptive statistics; after studies are retrieved and screened, users read PDFs of study content and feed extracted data directly into a database, which is used to produce "living" summary and interpretive statistical outputs. This platform has provided a basis for experimentation with the streamlining of data extraction, the end goal of which is automated identification, parsing, and abstraction of summary statistics reported in medical manuscripts.

Automated data extraction efforts

To begin to solve the problem of automated data extraction, it is first essential to understand the format in which input data is available. PDF manuscripts are the de-facto publication media. Within these manuscripts, the key data regarding the patient population/characteristics, interventions of interest, and outcomes are presented in both the text and data tables. Notably, the majority of previous extraction efforts have focused on textual extraction [8], despite the varied presentation styles and unstructured nature of both contextual information and the data themselves.

Targeting extraction from tables

The authors hypothesized that data tables, as opposed to free text, make an ideal target for automated extraction based on the following traits:

- Data in tables are densely concentrated
- Tables are delimited and typically include structure and a standardized set of contexts not found in free text
- Tables often report statistics not mentioned in the free text (e.g. secondary outcomes),
- Tables consistently report data with higher precision and full information (e.g. dispersion measures and sample sizes)

Existing standards for tabular presentation

Journals and medical research bodies have published standards around how statistics and tables should be presented [11-13]. Common themes include:

- Inclusion of units for continuous data elements
- Standardization of statistic formatting (e.g. mean (SD))
- Reporting interventions as full names or standardized abbreviations
- Reporting sample sizes used in an analysis

Yet table formatting standards, both in terms of styling and content, vary between journals. This heterogeneity has been noted previously, and a software tool targeted towards authors [14], tableone, was created to enable harmonized generation of statistical analyses and tables directly from study results. However, tableone and tools like it are not yet widely adopted enough to meaningfully standardize reporting.

Classifying tabular reporting practices

Therefore, automated tabular extraction depends on understanding the variety of table structures—including the types and frequencies of variation from common formats—in medical manuscripts. A recent systematic review shows that while 14 independent automation projects focused on full-text extraction have been published [15], only one project focused extraction on tables [8]. Furthermore, while the results of this project were promising, achieving high accuracy in machine-learning-based extraction, neither it nor any other study to date has surveyed or classified the structure of tables presented in the clinical literature. In the present study, the authors focus on identifying the characteristics that are essential for the automation of extraction of descriptive statistics in tables. This can provide concrete structural characteristics to enable assessment of the generalizability of tabular presentation formats and support the future automation of extraction.

Methods

Sampling published comparative clinical published studies

A cross-sectional sample of clinical study publication records was generated. In brief, published studies tagged as RCTs, as indexed in PubMed, from the year 2019 were searched using the following term: "randomized controlled trial" [Publication Type] AND 2019/01/01:2020/01/01[dp]. Search results were exported from PubMed on August 9th, 2021, using the Entrez Application Program Interface (API) [16], imported to the AutoLit platform, and search results were randomly sampled by index, without replacement, using the R function `sample`, to select 100 records. Of these, all published studies were included except those that met the following exclusion criteria: Not clinical, not comparative (i.e., the publication does not compare outcomes between patient groups), not in English, protocol only, or no full text available.

Abstracting tables

Within this sample, table attributes were identified, classified, and summarized. The following

Text 1
concepts for data extraction are introduced:

Text 2
Table 1
: Defining tabular concepts

Concept	Definition	Example
Data Element	A characteristic or quality being measured	Mortality
Metric	A measured instance of a data element, a descriptive statistic	2 / 59 (3.4%)
Arm	A subset of an experiment's participants that are assigned a specific intervention	Placebo Group
Time Point	The point(s) in time in an experiment when measurement of data elements is performed	6-month follow-up
Measurement Context	The combination of a data element, arm, and time point in experimental reporting	Mortality in the placebo group at 6 months

Text 3
Data extraction necessarily has to address the attributes of measurement context in order to discern the meaning of a given metric. Thus, for each patient arm, the intervention and population size must be identified, and for each data element the unit of measurement and time points must be extracted. Furthermore, metrics must be parsed into their constituent statistics:

Text 4
- Continuous metrics: the measure of central tendency and the dispersion measure;
- Dichotomous metrics: subset, total population, and percentage - n/N (%)
- Categorical metrics: subset and total population

Text 5
Tables and their descriptions typically contain at least partial representations of metrics and their measurement contexts. Context can be assigned to dimensions (rows and columns) of the table. For example, Figure 1 displays a "2x1" context, meaning the rows of the table correspond to 2 nested pieces of context (arms nested in data elements) and the columns correspond to 1 piece of context (time points). For comparison, Figure 2 displays a "1x1" context, wherein the data elements are labelled on the rows (with corresponding statistical formats) and arms are defined in the columns (timepoints not presented).

Text 6
Table Classification

Text 7
Each included record underwent full text review and was tagged in AutoLit based on the attributes of (1) its table structure, (2) its measurement context, and (3) its metric information. All tables within a published study were considered for classification via the tagging hierarchy. Tags were assigned on a per-table basis within each record such that the tags described the attributes of its tables.

Text 8
Table Structure Reporting

Text 9
As defined below in Table 2, table attributes covered the reporting of baseline and outcome data and table orientation/pagination. If a table did not report any descriptive statistics concerning patient characteristics or outcomes, it was accordingly not tagged.

Text 10
Table 2
: Tagging hierarchy of table structural attributes

Classification	Description	Tags
Baseline Reported	Demographic or baseline characteristics of one or more arms in the study are	**Tag** / **Applied When**: In Table / The study reports baseline characteristics for the study population in a table, broken out by arm

Table 1

	reported.	Participant-level	The study reports baseline characteristics for the study population in a table, reported for each participant
		No Arm-level Breakout	The study reports baseline characteristics for the entire study population in a table, with no break-out
Outcomes Reported	Experimental outcomes are reported.	**Tag**	**Applied When**
		In Table	The study reports outcomes in a table, including the primary outcome(s), at one or more follow-up time points
		Participant-level	The study reports outcomes in a table, including the primary outcome(s), at one or more follow-up time points, reported for each participant
		Secondary Only, In Table	The table reports outcomes in a table, but not all the primary outcomes of the study, instead focusing on datapoints other than primary outcomes.
Other Table Features		**Tag**	**Applied When**
		Rotated 90 Degrees	One or more baseline or outcome tables in the study is rotated 90 degrees in either direction on the page but is otherwise normal
		Multi-page	One or more baseline or outcome tables in the study overflows beyond its starting page but is otherwise normal

Text 1

Measurement Context Reporting

Text 2

Measurement context tags were applied for all tables for which patient baseline characteristics or outcomes were reported. Contextual information of interest included the pieces of context per array (where an array represents either a row or column), including data element headers, arm names and arm size reporting, labelling of interventions, and labelling of time points.

Text 3

3: Tagging hierarchy of measurement context attributes

Table 2

Classification	Description	Tags	
Table Context	The alignment of table dimensions and measurement context. Context is tagged as "Embedded" when individual header cells include 2 elements of context (e.g. "Baseline BMI").	**Tag**	**Applied When**
		1x1	Only two pieces of context are shown in the table dimensions: one on rows and one on columns. For example, data elements on rows and arms columns.
		2x1	Three pieces of context are shown in the table dimensions: two on rows and one on columns. For example, arms nested in data elements on rows, time points on columns.
		1x2	Three pieces of context are shown in the table dimensions: one on rows and two on columns.
Arm Size	The number of		

Table 1

Reported	participants in individual arms of the study	**Tag** / **Applied When** Embedded in Arm: Arm sizes are reported as part of the arm or intervention label (e.g. "Placebo (n=25)") Separate Array: Arm sizes are reported in a distinct column or row in the table
Intervention Labels	Header labels corresponding to the intervention(s) applied in each arm of the study	**Tag** / **Applied When** Full Name: The entire name of the intervention(s) for the arm is shown in the arm header Acronym or Abbreviated: An acronym or shortened version of the invention name(s) for the arm is shown in the arm header Control/ Experimental: The arm header is labelled with "Control" and "Experimental" or "Treatment" or "Intervention" Alternate Labels: Any header labeling scheme not identified above is used.
Time Point Labels	Header labels corresponding to the time at which the reported data was collected	**Tag** / **Applied When** Contains Unit of Time: The time point header contains an amount of time, including units. Pre/Post: The time point header is labelled "Pre"/"Post", "Before"/"After", "Baseline"/"Follow up" Incremental Numbered: Time point headers are labelled with numbers or letters in order of time (e.g. "t_1", "t_2")

Text 1

Metric Reporting

Text 2

Unlike table structure and measurement context, Metric tags were applied per-study rather than per-table. All baseline and outcome tables in the study were considered for metric classification.

Text 3

4: Tagging hierarchy of metric attributes

Table 2

Classification	Description	Tags
Statistic Format Reported	The format of statistics reported in a metric is displayed in the table or its description or footnotes. If multiple cases apply, the lowest in the table is the classification.	**Tag** / **Applied When** In Header: The statistic format, or just constituents are reported in the header of the array of metrics In Description or Footnotes: The statistic format or constituents are reported in the description or footnotes; these may apply to the entire table, or be annotations for arrays
Units Reported	The unit of a continuous data element; If multiple cases apply, the lowest in the table is the classification. If	**Tag** / **Applied When** In Header: The units of data elements are reported in each array header. In Description or Footnotes: The units of continuous data elements are reported in the description or footnotes; these may apply to the entire

Table 1

	units are missing on one or more data elements, this classification should be left empty.	<table><tr><td></td><td>table, or be annotations for arrays</td></tr><tr><td>Not Relevant</td><td>The study includes no continuous data elements, or the continuous data elements are unitless (e.g. scale data)</td></tr></table>
Statistic Formats	The pattern defining how the constituent statistics in array of metrics are formatted . The formats under each tag are created as they are encountered in studies.	<table><tr><th>Tag</th><th>Applied When</th></tr><tr><td>Continuous</td><td>The format is used for continuous data elements</td></tr><tr><td>Dichotomous</td><td>The format is used for dichotomous data elements; specifically, when only a single category is implied (e.g. “Mortality” or “Gender Male”)</td></tr><tr><td>Categorical</td><td>The format is used for categorical data elements; also applies when a dichotomous data element explicitly lists all categories (e.g. “Smoking” has separate arrays for “Yes” and “No”)</td></tr></table>
Categorical Categories	The method of reporting category labels for categorical data elements. This classification is left empty in the event that no categorical data elements are reported.	<table><tr><th>Tag</th><th>Applied When</th></tr><tr><td>Separate Array</td><td>Category labels are in an entirely separate (delimited) array from the data element header array</td></tr><tr><td>Same Array</td><td>Category labels are in the data element header array, with no distinction from other data element labels</td></tr><tr><td>Same Array Indented</td><td>Category labels are in the data element header array, but are nested under the categorical data element header via white-space or list indentation</td></tr><tr><td>In Cell</td><td>Categories are all reported in the same cell (e.g. “Gender M/F” with metrics “11 / 9”)</td></tr></table>

Text 1

Statistical Analysis

As a pilot study, no power analysis was performed to identify an appropriate sample size. Sample size was estimated to restrict 95% confidence intervals on proportions to +/- 15% in width. Frequencies were compiled with boolean queries on tags in AutoLit’s Study Inspector. Boolean queries were run on tagged studies using the open source software `btriev` [17] in NodeJS. The counts of results were then compiled and proportions confidence intervals were generated using a normal approximation with the `prop.test` function in R.Inferential statistics on proportions were built with a normal approximation and computed in the R programming language. Confidence intervals are reported at the 95% level.

Text 2

Results

Text 3

Characteristics of sampled studies

Text 4

Of the sampled 100 records, 12 published studies were excluded for lack of PDF full-text, 7 presented non-clinical findings, and 3 were not available in English, leaving 78 studies that were included in this pilot study. A single clinical study was reported in all (78 / 78, 100%) published manuscripts (referred to here as a ‘study’). Studies were published in 65 distinct journals, with the

Text 1
most frequent journal, PLoS One, publishing 9 of the studies. An interactive visualization of all studies tagged using the hierarchical paradigm described above is available on the Nested Knowledge Synthesis page (see Figure 3)[18].

Text 2
Reporting of baseline and outcome measures in tables

Text 3
Baseline and outcome data were reported in 71 of 78 (92%) studies (95% CI [Confidence Interval]: 84% - 96%) of studies. Both baseline and outcome data were presented in tables in 66 of 78 studies (85% [75% - 91%]), and 77 (99% [93% - 100%]) studies reported at least one table of baseline or outcome measures. Standard reporting (with arm-level breakout) was present in 97% ([90% - 99%]) of tables reporting baseline characteristics and 97% ([92% - 99%]) of tables reporting outcomes (Table 5).

Text 4
Table 1
5: Baseline and outcome reporting per study and per table

Type	Frequency per Study	Frequency per Table
Baseline Reported in Table	66 / 78 (85%)	67 / 174 (39%)
Arm-level breakout	64 / 66 (97%)	65 / 67 (97%)
No arm-level breakout	2 / 66 (3.0%)	2 / 67 (3.0%)
Outcomes Reported in Table	72 / 78 (92%)	107 / 174 (61%)
Arm-level breakout	69 / 72 (96%)	104 / 107 (97%)
Secondary Only	2 / 72 (2.8%)	2 / 107 (1.9%)
Participant-level	1 / 72 (1.4%)	1 / 107 (.93%)

Text 5
Among the 174 tables that were found to report either baseline or outcome descriptive statistics, 6 (3.4% [1.6% - 7.3%]) were rotated 90 degrees and 5 (2.9% [1.2% - 6.5%]) were multi-page.

Text 6
Reporting of measurement context in tables

Text 7
Results are shown with a denominator of the number of studies reporting one or more baseline or outcomes tables. 48 (62% [51% - 72%]) studies labelled arms by the full intervention name or an abbreviation of it. 22 of the 52 (42% [30% - 56%]) studies reporting time point context in tables labelled time points with an amount of time (Table 6).

Text 8
Table 2
6: Measurement context reporting per study

Type	Frequency per Relevant* Study
Arm Size Reported	57 / 77 (74%)
Embedded in Arm	50 / 77 (65%)
In Separate Array	6 / 77 (7.8%)
In Description	1 / 77 (1.3%)
Intervention Labels	77 / 77 (100%)
Control / Experimental	26 / 77 (34%)
Acronyms or Abbreviated	25 / 77 (33%)
Full Name	23 / 77 (30%)
Alternate Labels	3 / 77 (3.9%)
Time Point Labels	52 / 52 (100%)
Pre/Post	24 / 52 (46%)
Contains Unit of Time	22 / 52 (42%)
Incremental Numbering	6 / 52 (12%)

* One study reported no baseline or outcome data in tables and was thus left out from the

Text 1

measurement context analysis

Text 2

In terms of dimensions, the 1x1 context of data elements and arms was most commonly used to report findings, with time points included in only 4 of 99 (4% [1.6% - 9.9%]) these tables. Across all context types, arms were most frequently reported on columns (134 / 174; 77%), data elements on rows (144 / 174; 83%), and time points on columns (38 / 174; 22%; see Table 7).

Text 3

Table 1

7: Dimensions of tabular reporting of measurement context

Context dimensions	Frequency per Table
1x1	99 / 174 (57%)
DEs on rows, Arms on columns	90 / 99 (91%)
Arms on rows, DEs on columns	5 / 99 (5.1%)
Arm on rows, TP on columns	2 / 99 (2.0%)
TPs on rows, Arms on columns	2 / 99 (2.0%)
2x1	34 / 174 (20%)
TPs nested in DEs on rows, Arms on columns	18 / 34 (53%)
Arms Nested in DEs on Rows, TPs on columns	8 / 34 (24%)
DEs Nested in Arms on Rows, TPs on columns	2 / 34 (5.9%)
Arms Nested in TPs on Rows, DEs on columns	2 / 34 (5.9%)
DEs and TPs Embedded in Rows, Arms on columns	2 / 34 (5.9%)
TPs Nested in Arms on Rows, DEs on columns	1 / 34 (2.9%)
DEs Nested in TPs on Rows, Arms on columns	1 / 34 (2.9%)
1x2	26 / 174 (15%)
DEs on rows, TPs Nested in Arms on columns	16 / 26 (62%)
DEs on rows, Arms Nested in TPs on columns	5 / 26 (19%)
Arms on rows, TPs Nested in DEs on columns	3 / 26 (12%)
DEs on rows, Arms and TPs Embedded on columns	2 / 26 (7.7%)
Other Structure	15 / 174 (8.6%)
Stratified Reporting	8 / 15 (53%)
Only Reports Comparative Statistics	7 / 15 (47%)

DE - Data element, TP - Time point

Text 4

Reporting of metrics in tables

Text 5

75 (97%) of studies reported one or more continuous metrics in tables. 14 total formats were observed. Mean central tendencies were most frequently reported Mean ± SD (59%) and Mean (SD) (36%), while medians were most commonly reported as Median (IQR_1 - IQR_2) (52%; see Table 8). Seven formats were used for means, and 7 for medians.

Text 6

Table 2

8: Continuous metric reporting format in tables

Continuous Metrics Reported	Frequency per Relevant* Study
Mean	69 / 77 (90%)
Mean ± SD	41 / 69 (59%)
Mean (SD)	25 / 69 (36%)
Mean & SD in Separate Arrays	2 / 69 (2.9%)
Mean	2 / 69 (2.9%)
Mean SD	1 / 69 (1.4%)
Mean (CI_1 - CI_2)	1 / 69 (1.4%)
Mean ± SD (R_1 - R_2)	1 / 69 (1.4%)
Median	21 / 77 (27%)

Table 1

Median (IQR_1 - IQR_2)	11 / 21 (52%)
Median (R_1 - R_2)	4 / 21 (19%)
Median [IQR_1 - IQR_2]	3 / 21 (14%)
Median [IQR_1, IQR_2]	1 / 21 (4.8%)
Median (IQR_1 to IQR_2)	1 / 21 (4.8%)
Median (IQR*)	1 / 21 (4.8%)
R_1 - R_2 (Median)	1 / 21 (4.8%)

* One study reported no baseline or outcome data in tables and was thus left out from continuous data characterization

Text 1
Among 75 studies reporting at least one continuous metric, 64 (85% [76% - 92%]) contained data elements where units were relevant; of these, 55 (86% [75% - 92%]) reported units of measurement in table headers.

Text 2
Regarding dichotomous and categorical metrics, 8 different formats were encountered across 62 studies (Table 9); the format n (%) was most commonly observed for both metric types.

Text 3
Statistical formats (such as those shown in Table 9) were commonly represented in tables as well. In 65 of 77 studies (84%), the generalizable format was given. In 35 studies the format was presented in the header, while in 30 studies it was presented in the footnote or tabular description.

Text 4
9: Dichotomous and categorical formats and labels in tables

Table 2

Statistics Reported	Frequency per Relevant* Study
Dichotomous	40 / 77 (52%)
n (%)	30 / 40 (75%)
n	3 / 40 (7.5%)
n / (N - n)	3 / 40 (7.5%)
%	3 / 40 (7.5%)
n, %	2 / 40 (5.0%)
n & % in Separate Arrays	1 / 40 (2.5%)
Categorical	47 / 77 (61%)
n (%)	40 / 47 (85%)
n	8 / 47 (17%)
Categorical Labels	47 / 77 (61%)
Same Array, Indented	35 / 47 (74%)
Separate Array	7 / 47 (15%)
In Cell	7 / 47 (15%)
Same Array, Unindented	1 / 47 (2.1%)

* One study reported no baseline or outcome data in tables and was thus left out from dichotomous/categorical data characterization

Text 5
Among the 47 studies reporting categorical metrics, category label indentation under the data element header was observed in 35 (74% [60% - 85%]) studies. 2 studies used 2 distinct methods of category labelling.

Text 6
Across all data types, statistic formats were reported in tables or their descriptions in 84% ([75% - 91%]) of studies.

Text 7
Characteristics of studies with high information density

Text 8
Notwithstanding the variety of measurement contexts (with 15 independent formats detected) and statistic formats presented, high information density was common in our dataset. Studies that were

found to present maximal information in tables have the following classifications:

- Baseline Reported in Table or Outcomes Reported in Table
- Arm Size Reported
- Intervention Labels should be Full Name or Acronyms or Abbreviations
- When reporting relevant metrics, units of measurement are reported
- Statistic Format Reported

31 of 78 sampled studies (40% [30% - 51%]) match these classifications. The most impactful constraint among these classifications is Intervention Labels; if this classification is dropped, 48 (62% [50% - 75%]) of studies match the maximal-density, most-common formats listed above.

Discussion

Principal Results

In this pilot survey, we found that 85% of studies report both baseline and outcome data in tables, and 99% of studies have at least one table of baseline or outcome data. Arm-level (intervention-specific) data were presented in 97% of these tables, but there was major heterogeneity in the methods of reporting population size and intervention group names. Tabular dimensions ranged widely, with fifteen independent dimensional structures used to report measurement context across 174 tables. While 1x1 contexts represented the majority of tables, our results suggest automated context detection will need to contend with diversity of arrangements.

Similarly, while continuous data were very commonly reported (90% of studies) and dichotomous and categorical data were consistently reported (52% and 61% of studies, respectively), statistic formats were heterogeneous, with seven formats for means, seven for medians, and six for dichotomous data. Despite the heterogeneity of format, tables provided a consistent, high-density source for baseline and outcome data, and the contexts and formats defined here can be used to refine the expected data presentation for tabular data extraction. We plan to expand upon this pilot study in tabular structure with an expanded review-and-tabular-extraction study, wherein the framework outlined here will be used to classify and extract from underlying studies and the accuracy of extracted metrics will be determined by comparison to manual extraction.

Automated recognition of context and format

The context presented in tables showed the most disappointing rate of reporting information of interest. While arm sizes were reported in tables in 70% of publications, arm interventions and measurement time points were reported in a self-contained manner 60% and 40% of the time, respectively. If tables are extracted in a completely self-contained manner, with no access to the publication's full text, we expect only 40% of publications will contain sufficient information in tables alone to complete extraction. Extraction automations will therefore have to receive human input or consult the free text to supplement arm and time point context from tables.

Statistic format reporting was heterogenous in format but extremely commonly reported: statistic format was explicitly reported in tables in up to 90% of publications. Even where absent, the format may also be inferred from the metric arrays themselves. Inferred formats may be useful when formats are not reported, or as a validation on the detected format. Categorical metrics may produce the most complexity for automations, as category labels are often not distinguished from data element labels by more than whitespace indentation.

Given the commonality of n (%) reporting for dichotomous and categorical data elements, it may be possible to arithmetically derive missing arm sizes from metrics. If not essential, mining the full text

may still provide value in validations or completing data. For example, interventions were reported as abbreviations or acronyms in around 30% of publications; pattern matching on the abstract or introduction could generate full strings for these shortened versions.

A non-trivial number of publications, around 5%, contained rotated or multi-page tables. Automations should consider tools to identify and apply corrective measures in these cases. Eight percent of tables reported only stratified data or only comparative statistics; while these cases are typically mathematically correctable to arm-level data that meta-analysts desire, automation of these procedures would add complexity. Lastly, while no publication included in our pilot study had missing data, potential missing data must be addressed in any automated workflow: we suggest that for any table where data were missing, the table should be visually presented to users for confirmation.

Previous research on tabular extraction

To date, the scientific literature does not seem to contain any studies giving an overview of the table structure, context presentation, and statistic formats, making this pilot study the first of its kind. The Cochrane Collaboration has created a test dataset for automated extraction that may be used to test the accuracy of novel extraction algorithms, but their dataset did not classify tabular structure, instead focusing on providing the test/training datasets and preliminary testing of their own semi-automated extraction system [19].

However, previous authors have proceeded beyond classification and provided approaches for automating tabular extraction. Unlike the approaches reviewed by Jonnalagadda et al. [8] and Marshall et al. [20], which focus on simplified content-extraction tasks from free-text (such as abstracts), Milosevic et al. actually tested a preliminary algorithm for tabular extraction [21]. While this study did not include an overview of the context or statistic formats, the authors achieved an F-score (accuracy) of 82%-92% in content extraction from a simplified set of HTML tables with 1x1 context formats. The seven-step process for detecting, processing, tagging, and extracting from tables used by Milosevic et al. is the most complete tabular extraction process published to date. The only competing approach focused on tables in medical publications was from Xu et al. [22], who were able to extract drug-side effect relationships with 52% accuracy using a statistical classifier. Other than Milosevic et al.'s pilot study, then, despite at least 26 approaches attempted in textual extraction [8], automated extraction remains an unmet need for which tabular extraction is a promising and under-examined methodology.

Limitations

We believe our findings will generalize to modern clinical publications due to the simplicity of our search and applicability of classifications. However, this survey and the AutoLit data extraction framework are applicable only to clinical research publications. Since our search was limited to RCTs, some study types, such as observational studies, may show different characteristics. Similarly, we did not stratify our results by journal, impact factor, or other factors apart from filtering to RCTs,

Text 1
so journal-related characteristics may affect how representative this pilot is of medical publishing generally.

Text 2
Additionally, specific fields of research may show characteristics that do not align well with the averaging-across-fields approach used in our study. As a pilot survey, our study did not involve a power analysis; however, this pilot study can be used to determine sample sizes quantitatively in future research. Lastly, our breakout of contexts and formats is always subject to expansion not covered in this sample, and automations built on the expectation of a limited set of contexts or formats may fail when new presentations of this information are encountered. The test of this framework will be the accuracy of extraction algorithms that employ it, compared against existing extraction methods.

Text 3
Conclusions

Text 4
In this pilot survey, we found a high density of information in tables, with over 85% of studies reporting both background and outcome measures in tables, but with major heterogeneity in presentation of measurement context. Measurement context was most often presented in 1x1 format, but fifteen independent formats were found. Similarly, means and medians were each found in seven independent formats, and dichotomous variables in six. Despite this, high-quality contextual information (intervention labels, arm sizes, units, and statistic formats) were presented in 40% of studies. The range of context and statistic formats surveyed here can provide a baseline for future tabular extraction efforts.

Text 5
Acknowledgements

Text 6
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Text 7
Conflicts of Interest

Text 8
Karl Holub works for and holds equity in Nested Knowledge, Inc. Nicole Hardy works for and holds equity in Nested Knowledge, Inc. Kevin Kallmes works for and holds equity in Nested Knowledge, Inc., and holds equity in Superior Medical Experts, Inc.

Text 9
Abbreviations

Text 10
JMIR: Journal of Medical Internet Research
RCT: Randomized controlled trial
DE: Data element
TP: Time point
SD: Standard deviation
CI: Confidence Interval
IQR: Interquartile-range
R: Range
SR/MA: Systematic review and meta-analysis
PDF: Portable document format
API: Application program interface
BMI: Body mass index

Figure Captions

Figure 1. An example of a 2x1 context with arms (blue) nested in data elements (red) on rows and time points (green) on the columns. Table from Chellappa et al. [23].

Figure 2. A 1x1 baseline table reporting data elements on rows and arms on columns. Arm sizes are embedded in intervention headers (red), category labels are reported in the data element array indented (blue), and statistic formats are reported in headers (green). Table from Benitez et al. [24].

Figure 3. A screenshot of the interactive tagging hierarchy applied across the 78 studies included in this pilot survey. Two filters are applied: “2x1” was selected first, and then “Mean ± SD,” meaning the sunburst plot is filtered to studies for which both tags are present. The right menu displays the 15 studies for which this is true as well as statistics about how common the tags in question are across all included studies.

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Text 1

Supplementary Files

Preprint
JMIR Publications

Figures

Preprint JMIR Publications